Chapter 1

Types of prime power cyclotomic cosets

In this report, we consider prime power p^k-cyclotomic cosets instead of p^3. Since in p^2, cyclotomic cosets are of 2 types and p^3 are of 3. So p^k will be of k types. Instead of saying types I, II, III, IV, ..., it is better to say type 1, 2, 3, 4, ..., k but we will start from type 0 to $k-1$ for some purpose which will be explained later in this chapter.

Definition 1. Let m and k be positive integers, p be a prime number and T be an integer such that $\gcd(m, p) = 1$ and $0 \leq T < k$. For each $a \in \mathbb{Z}_m$, the p^k-cyclotomic coset of $\mathbb{Z}_m$ containing a, $S_{p^k}(a)$, is said to be of **type** T if $S_{p^k}(a) = S_{p^k}(-p^T a)$ or **type** T' if $S_{p^k}(a) \neq S_{p^k}(-p^T a)$.

Note that if we define $S_{p^k}(a)$ to be of type k, it is exactly the same to type 0 , type $k+1$ is type 1 and so on. Hence it is no need to define type that is greater than $k-1$. When $k = 3$, type I, II and III in p^k become type 0, 1 and 2 respectively. From now on, without causing a confusion, we denote $S_{p^k}(a)$ by $S(a)$.

When we are going to retrieve those k types of $S_{p^k}(a)$, we can reduce our work by calculating on some types to imply others by the following theorems.

Theorem 2. Let m and k be positive integers, p be a prime number and T be an integer such that $\gcd(m, p) = 1$ and $0 < T < k$. For each $a \in \mathbb{Z}_m$, $S_{p^k}(a)$ is of type T if and only if it is of type $k - T$.

Proof. Suppose $S(a)$ is of type $T \neq 0$, i.e., $S(a) = S(-p^T a)$. Then there exists an integer s such that $-p^T a = p^{ks} a$. Multiplying both side by $-p^{k-T}$ gives $p^k a = -p^{ks+k-T} a = p^{ks} \cdot -p^{k-T} a$. Thus $S(a) = S(-p^{k-T} a)$, that is, $S(a)$ is of type $k - T$. Conversely, suppose $S(a)$ is of type $k - T$, i.e., $S(a) = S(-p^{k-T} a)$. Then there exists an integer s such that $-p^{k-T} a = p^{ks} a$. Multiplying both side by $-p^T$ gives $p^k a = -p^{ks+T} a = p^{ks} \cdot -p^T a$. Thus $S(a) = S(-p^T a)$, that is, $S(a)$ is of type T. □

Corollary 3. Let m and k be positive integers, p be a prime number and T be an integer such that $\gcd(m, p) = 1$ and $0 < T < k$. For each $a \in \mathbb{Z}_m$, $S_{p^k}(a)$ is of type T' if and only if it is of type $(k - T)'$.

Example 4. In $\mathbb{Z}_{45}$, $S_{2^9}(5)$ is of type $0, 1', 2', 3, 4'$, so it is of type $8', 7', 6, 5'$.

Calculating types is, sometimes, quite difficult when k is large. So we define the following set to reduce the process.

Definition 5. Let k be a positive integer. For each $T \in \mathbb{Z}_k$, we denote **the set of consequent types of** T **in power** k by $C_k(T) := \{2Ti + T \mid i \in \{0, 1, \dots\}\}$.

For a type T from now on, when $\mathrm{ord}(T)$ is mentioned we mean the additive order of T in $\mathbb{Z}_k$ not of $\mathbb{Z}_m$.

Theorem 6. Let k be a positive integer. For each $T \in \mathbb{Z}_k$, let O be an additive order of T. We have that

1. If $2 \nmid O$, then
$$0, T, 2T, \dots, (O - 1)T$$
are exactly O distinct elements in $C_k(T)$.

2. If $2|O$ then
$$T, 3T, \dots, (O - 1)T$$
are exactly $\frac{O}{2}$ distinct elements in $C_k(T)$.

Thus the number of distinct elements in $C_k(T)$ can be simplified by,

$$|C_k(T)| = \frac{k}{\gcd(k, 2\gcd(T,k))}.$$

Proof. 1. Suppose $2 \nmid O$. We have that k is odd and

$$C_k(T) = \{T, 3T, \ldots, kT = 0, 2T, 4T, \ldots, (O-1)T, (O+1)T = T\}.$$

Since $\mathrm{ord}(T) = O$, by the definition of order, $0, T, \ldots, (O-1)T$ are distinct.

2. Suppose $2|O$. We have

$$C_k(T) = \{T, 3T, \ldots, (O-1)T, (O+1)T = T\}.$$

We have $T, 3T, \ldots, (O-1)T$ are distinct. Thus $C_k(T)$ has $\frac{O}{2}$ distinct elements as we described.

To simplify the order of $C_k(T)$, we have already shown that $|C_k(T)| = \frac{O}{\gcd(O,2)}$. From $O = \frac{k}{\gcd(T,k)}$, substitution gives

$$\begin{aligned} |C_k(T)| &= \frac{k}{\gcd(T,k)\cdot\gcd(\frac{k}{\gcd(T,k)}, 2)} \\ &= \frac{k}{\gcd(\gcd(T,k)\cdot\frac{k}{\gcd(T,k)}, \gcd(T,k)\cdot 2)} \\ &= \frac{k}{\gcd(k, 2\gcd(T,k))} \end{aligned}$$

and we are done. □

Theorem 7. Let m and k be positive integers, p be a prime number and T be an integer such that $\gcd(m,p) = 1$ and $0 \le T < k$. For each $a \in \mathbb{Z}_m$, if $S_{p^k}(a)$ is of type T then it is also of $\frac{k}{\gcd(k,2\gcd(T,k))}$ types in $C_k(T)$.

Proof. Suppose $S(a)$ is of type T, i.e., $S(a) = S(-p^T a)$. Then there exists an integer s such that $p^{ks}a = -p^T a$. Multiplying both side by $-p^T$ gives $p^{ks}\cdot -p^T a = p^{2T}a$ which is equivalent to $S(p^{2T}a) = S(-p^T a) = S(a)$. Multiplying both side by $-p^T$ again gives $p^{ks}\cdot p^{2T}a = -p^{3T}a$ which is equivalent to

$$S(-p^{3T}a) = S(p^{2T}a) = S(-p^T a) = S(a).$$

By repeating this process, we finally have

$$S(a) = S(-p^T a) = S(-p^{3T} a) = S(-p^{5T} a) = \dots.$$

Thus it is of type $2Ti + T \bmod k$, which are in the set of consequent types of T in power k, $C_k(T)$. $\square$

Note that the consequent types of $S_{p^k}(a)$ is independent to the prime p and the integer m but depends on the type T and power of the prime k.

Remark 8. Let k be a positive integer. For each $T \in \mathbb{Z}_k$, $C_k(T) = C_k(k - T)$.

The converse of the theorem need not be hold by the following example.

Example 9. From $C_9(1) = \{0, 1, \dots, 8\}$. In $\mathbb{Z}_{81}$, $S_{2^9}(27)$ is of type 1 so it is of types $0, 1, \dots, 8$, while $S_{2^9}(3)$ is of type $0 \in C_9(1)$ but it is not of type 1 and any other types.

Corollary 10. Let m be a positive integer and p, q be prime numbers such that $\gcd(m, p) = 1$ and q is odd. For each $a \in \mathbb{Z}_m$, $S_{p^q}(a)$ has the same being of type $1, 2, \dots, q - 1$.

Proof. Suppose $S(a)$ is of type $T > 0$. Then it is of types in $C_q(T)$. Since $0 < T < q$ and q is prime, we have $O := \operatorname{ord}(T) = q$. Thus $2 \nmid O$ and $C_q(T) = \{0, T, 2T, \dots, (q - 1)T\}$ is the set of q distinct elements in $\mathbb{Z}_q$. Hence $C_q(T) = \{0, 1, 2, \dots, q - 1\}$. Therefore it is of type $1, 2, \dots, q - 1$. Conversely, suppose $S(a)$ is not of type $T > 0$. We will show that it is not of type $1, 2, \dots, q - 1$. Suppose it is of some type $U > 0$. The first part of this proof implies $S(a)$ is of type T, which contradicts the assumption. $\square$

Note that type 0 is not included since it is the only type having an additive order 1 for all k.

Example 11. $S_{2^{11}}(a)$ has the same being of type $1, \dots 10$ in $\mathbb{Z}_{45}$, in fact, for every $\mathbb{Z}_m$ that $11 \nmid m$.

Corollary 12. Let m and k be positive integers, p be a prime number and T be an integer such that k is odd, $\gcd(m,p) = 1$ and $0 \leq T < k$. For each $a \in \mathbb{Z}_m$, if $\mathrm{S}_{p^k}(a)$ is of type T then it is of type 0.

Proof. Suppose $2 \nmid k$. Since $\mathrm{ord}(T)|k$, we have that $2 \nmid \mathrm{ord}(T)$. Then $0 \in C_k(T)$. Hence $S(a)$ is of type 0. $\square$

The next one to define is a function λ which is χ and λ in the last project. Since we cannot define new functions by greek letters forever. So, for convenient, we merge those functions together by preserving their definitions.

Definition 13. Let k be a positive integer, p be a prime number and T be an integer such that $0 \leq T < k$, let $\lambda_{p^k,T} : \mathbb{N} \to \{0,1\}$ be a function defined by

$$\lambda_{p^k,T}(q) = \begin{cases} 0 & \text{, if there exists an integer } s \geq 0 \text{ such that } q \mid p^{ks+T}+1, \\ 1 & \text{, otherwise} \end{cases}$$

for all $q \in \mathbb{N}$.

Next we will show how this λ can be applied.

Theorem 14. Let m and k be positive integers, p be a prime number and T be an integer such that k is odd, $\gcd(m,p) = 1$ and $0 \leq T < k$. For each $a \in \mathbb{Z}_m$, $S_{p^k}(a)$ is of type T if and only if $\lambda_{p^k,T}(\mathrm{ord}(a)) = 0$.

Proof. Suppose that $S(a)$ is of type T. Then $S(a) = S(-p^T a)$. Which is equivalent to $a \equiv p^{ks}(-p^T a) \mod m$ for some integer s. Therefore

$$\begin{aligned} 0 &\equiv (p^{ks+T}+1)a && \mod m \\ &\equiv (p^{ks+T}+1) && \mod \mathrm{ord(a)} \end{aligned}$$

Thus $\lambda_{p^k,T}(\mathrm{ord}(a)) = 0$. Conversely, suppose that $\lambda_{p^k,T}(\mathrm{ord}(a)) = 0$. Then $\mathrm{ord}(a)|p^{ks+T}+1$ for some integer s. Which is equivalent to

$$\begin{aligned} 0 &\equiv p^{ks+T}+1 && \mod \mathrm{ord(a)} \\ &\equiv (p^{ks} \cdot p^T + 1)a && \mod a\mathrm{ord}(a) \\ &\equiv (p^{ks} \cdot p^T + 1)a && \mod m \\ p^{ks} \cdot -p^T a &\equiv a && \mod m \end{aligned}$$

since $a\mathrm{ord}(a)$ is multiple of m. Thus $S_q(a) = S_q(-p^T a)$. Therefore $S_q(a)$ is of type T. $\square$

Note that when we apply this function to the cyclotomic cosets, we have that $\gcd(p, q) = 1$.

Example 15. By calculating, $\lambda_{3^8,T}(\mathrm{ord}(0)) = \lambda_{3^8,T}(1) = 0$ for all T. Then $S_{p^k}(0)$ is of all type. In the other hand, $S_{3^8}(10)$ is of type $0', 1, 2', 3, 4', 5, 6'$ and 7 in $\mathbb{Z}_{70}$ so $\lambda_{3^8,T}(\mathrm{ord}(10)) = 0$ for $T \in \{1, 3, 5, 7\}$ and $\lambda_{3^8,T}(\mathrm{ord}(10)) = 1$ for $T \in \{0, 2, 4, 6\}$.

Theorem 16. Let a be a positive integer, p and k be prime numbers where k is odd. If $\gcd(a, p) = 1$, the following $k - 1$ statements are equivalent:

- $a \mid p^{ks+1} + 1$ for some $s \geq 0$.
- $a \mid p^{ks+2} + 1$ for some $s \geq 0$.

  ...
- $a \mid p^{ks+k-1} + 1$ for some $s \geq 0$.

Proof. For $T \in \{1, 2, \ldots, k - 1\}$, let $P(T)$ be the statement

$$a \mid p^{ks+T} + 1 \text{ for some } s \geq 0.$$

$(P(T) \to P(T + 1))$ Suppose $a \mid p^{ks+T} + 1$ for some $s \geq 0$. Then $\lambda_{p^k,T}(a) = 0$ and $S_{p^k}(a)$ is of type T in $\mathbb{Z}_{a^2}$ since $\mathrm{ord}(a) = a$. Then we have $S(a)$ is of type $T + 1$, which implies $\lambda_{p^k,T+1}(a) = 0$, i.e., $a \mid p^{kr+T+1} + 1$ for some $r \geq 0$ and we are done. $(P(k - 1) \to P(1))$ Suppose $a \mid p^{ks+k-1} + 1$ for some $s \geq 0$. Then $\lambda_{p^k,k-1}(a) = 0$ and $S_{p^k}(a)$ is of type $k - 1$ in $\mathbb{Z}_{a^2}$ since $\mathrm{ord}(a) = a$. Thus it is of type 1, which implies $\lambda_{p^k,1}(a) = 0$, i.e., $a \mid p^{kr+1} + 1$ for some $r \geq 0$ and completes the proof. $\square$

Theorem 17. Let m and k be positive integers, p be a prime number and T be an integer such that k has a factor $\bar{k} \neq k$, $\gcd(m, p) = 1$ and $0 \leq T < k$. For each $a \in \mathbb{Z}_m$, if $S_{p^k}(a)$ is of type of $\bar{k}$ then $S_{p^q}(a)$ is of type 0 for all $q \mid \bar{k}$ and $q \mid k - \bar{k}$.

Proof. Since k is a positive integer, there exists different primes $k_1, k_2, \ldots, k_n$ and positive integers $l_1, l_2, \ldots, l_n$ such that

$$k = k_1^{l_1} k_2^{l_2} \ldots k_n^{l_n}.$$

Suppose $S_{p^k}(a)$ if of type $\bar{k} = k_1^{r_1} k_2^{r_2} \ldots k_n^{r_n} \neq k$ when $0 \leq r_i \leq l_i$ for all $i \in \{1, 2, \ldots, n\}$. Then $p^{kt}a = -p^{k_1^{r_1} k_2^{r_2} \ldots k_n^{r_n}} a$ for some integer t which is $-a = p^{kt - k_1^{r_1} k_2^{r_2} \ldots k_n^{r_n}} a = p^{k_1^{r_1} k_2^{r_2} \ldots k_n^{r_n} (\frac{k}{k_1^{r_1} k_2^{r_2} \ldots k_n^{r_n}} t - 1)} a$. From $k_1^{r_1} k_2^{r_2} \ldots k_n^{r_n}$ is a factor of k, $\frac{k}{k_1^{r_1} k_2^{r_2} \ldots k_n^{r_n}} t - 1$ is equal to some integer u so that $-a = p^{k_1^{r_1} k_2^{r_2} \ldots k_n^{r_n} u} a = (p^{\bar{k}})^u a$ which implies $S_{p^{\bar{k}}}(a) = S_{p^{\bar{k}}}(-a)$. Thus $S_{p^{\bar{k}}}(a)$ is of type 0. In addition, for all $q \mid \bar{k}$, we have $\frac{\bar{k}}{q}$ is equal to some integer v so that $-a = p^{(q\frac{\bar{k}}{q})u} a = p^{q(vu)} a$ which implies $S_{p^q}(a)$ is of type 0. The latter condition can be retrieved by the fact that $S_{p^k}(a)$ if of type $k - \bar{k}$. $\square$

Theorem 18. Let m and k be positive integers, p be a prime number, $a \in \mathbb{Z}_m$ and T be an integer such that $\gcd(m, p) = 1$, $S_{p^k}(a)$ is of type 0 and $0 < T < k$. Then $S_{p^k}(a)$ is of type T if $\mathrm{ord}_{\mathrm{ord}(a)} p \mid T$.

Proof. Let $S(a)$ be of type 0, that is, $\lambda_{p^k,0}(\mathrm{ord}(a)) = 0$ which is equivalent to $\mathrm{ord}(a) \mid p^{ks} + 1$, for some integer s. In addition, we have that $\mathrm{ord}(a) \mid (p^{ks} + 1)p^T$ and

$$(p^{ks} + 1)p^T = p^{ks+T} + p^T = (p^{ks+T} + 1) + (p^T - 1).$$

Suppose $\mathrm{ord}_{\mathrm{ord}(a)} p \mid T$. Then $p^T \equiv 1 \mod \mathrm{ord(a)}$ and $\mathrm{ord}(a) \mid (p^{ks} + 1)p^T - (p^T - 1)$ which is $\mathrm{ord}(a) \mid (p^{ks+T} + 1)$. Hence $\lambda_{p^k,T}(\mathrm{ord}(a)) = 0$ which implies $S(a)$ is of type T. $\square$

Chapter 2

Improvements of λ

We have been trying to solve the value of $\lambda_{p^k,T}$ in the simpliest way because finding s in its function is hard sometimes. So it will be discussed in this chapter. Recall that, for each prime p, positive integers k, q and integer T satisfying $\gcd(p, q) = 1$ and $0 < T < k$,

$$\lambda_{p^k,T}(q) := \begin{cases} 0 & \text{, if there exists an integer } s \geq 0 \text{ such that } q \mid p^{ks+T} + 1, \\ 1 & \text{, otherwise.} \end{cases}$$

We will begin studying λ when k is odd.

Theorem 19. Let p and q be positive integers. For an odd integer k, $q \mid p^{ks} + 1$ for some integer s if and only if $q \mid p^{lt} + 1$ for all odd integers l and some integer t.

Proof. It is enough to prove that, for a given odd integer k, $q \mid p^{ks} + 1$ for some integer s if and only if $q \mid p^{(k+2)t} + 1$ for some integer t. Suppose $q \mid p^{ks} + 1$ for some integer s. We have

$$\begin{aligned} p^{ks} &\equiv -1 && \mod q \\ p^{ks}p^{2s} &\equiv -p^{2s} && \mod q \\ (p^{(k+2)s})^k &\equiv (-p^{2s})^k && \mod q \\ p^{(k+2)(sk)} &\equiv (-1)^k(p^{2s})^k && \mod q \\ &\equiv -(p^{sk})^2 && \mod q \\ &\equiv -(-1)^2 && \mod q \\ &\equiv -1 && \mod q \end{aligned}$$

When we choose $t = sk$, we have $q \mid p^{(k+2)t} + 1$ as desired. Conversely, suppose $q \mid p^{(k+2)t} + 1$ for some integer t.

$$\begin{aligned} p^{(k+2)t} &\equiv -1 && \mod q \\ (p^{kt+2t})^k &\equiv (-1)^k && \mod q \\ p^{k(kt+2t)} &\equiv -1 && \mod q \end{aligned}$$

Choosing $s = kt + 2t$ gives $q \mid p^{ks} + 1$ and completes the proof. $\square$

This theorem gives us some results to λ.

Corollary 20. Let p be prime and k, l be odd, then $\lambda_{p^k,0}(q) = \lambda_{p^l,0}(q)$ for all integer q.

Corollary 21. Let a be a positive integer, p be a prime number and k, l be odd, then $S_{p^k}(a)$ is of type 0 if and only if $S_{p^l}(a)$ is of type 0.

Theorem 22. Let p be prime and k be odd, then $\lambda_{p^k,0}(1) = 0$, $\lambda_{p^k,0}(2) = 0$ for all prime $p > 2$ and

$$\lambda_{p^k,0}(o) = \begin{cases} 0 & , 2 \mid \mathrm{ord}_o(p) \text{ and } p^{\frac{\mathrm{ord}_o(p)}{2}} = -1, \\ 1 & , \text{otherwise} \end{cases}$$

for all $o > 2$. Note that the calculation is taken on $\mathbb{Z}_o$.

Proof. Clearly, for all integer k, $\lambda_{p^k,0}(1) = 0$ since $1 \mid p^k + 1$ for all integer p and $\lambda_{p^k,0}(2) = 0$ since $2 \mid p^k + 1$ for all odd prime p.

Suppose $2 \mid \mathrm{ord}_o(p)$ and $p^{\frac{\mathrm{ord}_o(p)}{2}} = -1$. Thus $(p^{\frac{\mathrm{ord}_o(p)}{2}})^k = (-1)^k = -1$ and $o \mid p^{k\frac{\mathrm{ord}_o(p)}{2}} + 1$ which is equivalent to $\lambda_{p^k,0}(o) = 0$.

On the other hand, suppose $2 \nmid \mathrm{ord}_o(p)$ or $p^{\frac{\mathrm{ord}_o(p)}{2}} \neq -1$. Then the solution of $p^x = -1$ does not exists. We will show a contradiction by suppose it does, choose the smallest $0 \leq x < \mathrm{ord}_o(p)$ satisfying the equation. If $2 \nmid \mathrm{ord}_o(p)$, then $p^{2x} = (p^x)^2 = (-1)^2 = 1$ implies $\mathrm{ord}_o(p) \mid 2x$ but $2x < 2\mathrm{ord}_o(p)$, we have that $2x = \mathrm{ord}_o(p)$ and hence $2 \mid \mathrm{ord}_o(p)$. If $p^{\frac{\mathrm{ord}_o(p)}{2}} \neq -1$. Then $p^{2x} = 1$ implies $\mathrm{ord}_o(p) \mid 2x$ thus $x = \frac{\mathrm{ord}_o(p)}{2}$, a contradiction. So $\mathrm{ord}(o) \nmid p^x + 1$ for all x implying $\lambda_{p^k,0}(o) = 1$.

□

Definition 23. For each positive integer k, let $u : \mathbb{N} \to \mathbb{N}$ be a function defined by $u(k) = \min\{s \mid 2^s \nmid k \text{ where } s > 0\}$. In fact, $u(k)$ is the smallest positive integer satisfying $2^{u(k)} \nmid k$.

Remark 24. $u(2^k) = k + 1$ for all integer k.

Theorem 25. Let k, l, p, q be positive integers such that $u(k) = u(l)$. Then $q \mid p^{ks} + 1$ for some integer s if and only if $q \mid p^{lt} + 1$ for some integer t.

Proof. Suppose $q \mid p^{ks} + 1$ for some integer s. Which is equivalent to

$$\begin{aligned} -1 &\equiv p^{ks} && \mod q \\ &\equiv (p^{\frac{k}{2^{u(k)-1}}s})^{2^{u(k)-1}} && \mod q \\ &\equiv (p^{\frac{l}{2^{u(l)-1}}t})^{2^{u(l)-1}} && \mod q \\ &\equiv p^{lt} && \mod q \end{aligned}$$

for some integer t since both $\frac{k}{2^{u(k)-1}}$ and $\frac{l}{2^{u(l)-1}}$ are odd. Equivalently, $q \mid p^{lt} + 1$ and we are done. □

Theorem 26. Let p be prime and k be a positive integer. If $\lambda_{p^k,0}(q) = 0$ then $\lambda_{p^l,0}(q) = 0$ for all integers $l > 0$ satisfying $u(l) < u(k)$.

Proof. Suppose $q \mid p^{ks} + 1$ for some integer s. Then, in modulo q,

$$-1 \equiv (p^{\frac{k}{2^{u(k)-1}}s})^{2^{u(k)-1}} \equiv (p^{mt})^{2^{u(k)-1}} \equiv (p^{mt})^{2^{u(l)-1+u(k)-u(l)}} \equiv p^{2^{u(l)-1}t \cdot 2^{u(k)-u(l)}m} \equiv p^{ln}$$

for some integer m, t where t is odd, $l = 2^{u(l)-1}t$ and $n = 2^{u(k)-u(l)}m$. □

Corollary 27. Let p be prime and k be a positive integer. If $\lambda_{p^l,0}(q) = 1$ then $\lambda_{p^k,0}(q) = 1$ for all integers $k > 0$ satisfying $u(k) > u(l)$.

Lemma 28. Let p be prime and k be a positive integer, then $\lambda_{p^k,0}(q) = \lambda_{p^{2^{u(k)-1}},0}(q)$ for all $q \in \mathbb{N}$.

Theorem 29. Let p be prime and k be a positive integer, then $\lambda_{p^k,0}(1) = 0$, $\lambda_{p^k,0}(2) = 0$ for all prime $p > 2$ and

$$\lambda_{p^k,0}(q) = \begin{cases} 0 & , 2^{u(k)} \mid \mathrm{ord}_q(p) \text{ and } p^{\frac{\mathrm{ord}_q(p)}{2}} = -1, \\ 1 & , \text{otherwise} \end{cases}$$

for all $q > 2$. Note that the calculation is taken on $\mathbb{Z}_q$.

Proof. The results for $q = 1$ and 2 has been proved. By previous theorem, it is equivalent to show that, for all $k \geq 0$,

$$\lambda_{p^{2^k},0}(q) = \begin{cases} 0 & , 2^{k+1} \mid \mathrm{ord}_q(p) \text{ and } p^{\frac{\mathrm{ord}_q(p)}{2}} = -1, \\ 1 & , \text{otherwise} \end{cases}$$

We will use mathematical induction on k. Case $k = 0$ has been proved since 2^k is odd. Suppose this formula holds for k, we will show that it also holds for $k + 1$.

Suppose $2^{k+2} \mid \mathrm{ord}_q(p)$ and $p^{\frac{\mathrm{ord}_q(p)}{2}} = -1$. We will show that $\lambda_{p^{2^{k+1}},0}(q) = 0$. Since $\mathrm{ord}_q(p) = 2^{k+2}m$ for some integer m. We have $-1 = p^{\frac{\mathrm{ord}_q(p)}{2}} = p^{\frac{2^{k+2}m}{2}} = p^{2^{k+1}m}$ which implies $q \mid p^{2^{k+1}m} + 1$ thus $\lambda_{p^{2^{k+1}},0}(q) = 0$ by its definition.

On the other hand, suppose $2^{k+2} \nmid \mathrm{ord}_q(p)$. If $\lambda_{p^{2^k},0}(q) = 1$ then $\lambda_{p^{2^{k+1}},0}(q) = 1$, so we consider in case of $\lambda_{p^{2^k},0}(q) = 0$, by induction hypothesis, $2^{k+1} \mid \mathrm{ord}_q(p)$ and $p^{\frac{\mathrm{ord}_q(p)}{2}} = -1$. Let m be an integer such that $\mathrm{ord}_q(p) = 2^{k+1}m$, because $2^{k+2} \nmid \mathrm{ord}_q(p)$, we have m is odd. By contradiction, suppose $\lambda_{p^{2^{k+1}},0}(q) = 0$, by the definition of λ, there exists some integer a_{k+1} such that $q \mid p^{2^{k+1}a_{k+1}} + 1$ then $p^{2^{k+1}a_{k+1}} = -1$ and $p^{2^{k+2}a_{k+1}} = 1$. Thus $\mathrm{ord}_q(p) \mid 2^{k+2}a_{k+1}$, which is, $2^{k+1}m \mid 2^{k+2}a_{k+1}$ implying $m \mid 2a_{k+1}$, but m is odd so $m \mid a_{k+1}$. Hence $-1 = p^{2^{k+1}a_{k+1}} = p^{2^{k+1}m\frac{a_{k+1}}{m}} = (p^{\mathrm{ord}_q(p)})^{\frac{a_{k+1}}{m}} = 1^{\frac{a_{k+1}}{m}} = 1$, a contradiction, $\lambda_{p^{2^{k+1}},0}(q) = 1$. Lastly, suppose $p^{\frac{\mathrm{ord}_q(p)}{2}} \neq -1$, hence $\lambda_{p^{2^k},0}(q) = 1$ by induction hypothesis which implies $\lambda_{p^{2^{k+1}},0}(q) = 1$.

Therefore $\lambda_{p^{2^k},0}(q)$ of this formula holds for all integer $k \geq 0$. $\square$

All of theorems we have proved are all about when $T = 0$. Next, we will consider when $T = 1$ and $k \geq 2$.

Theorem 30. Let p and k be prime, then for all integer r, $q \mid p^{k^r s+1}+1$ for some integer s if and only if $q \mid p^{kt+1}+1$ for some integer t.

Proof. Suppose $q \mid p^{k^r s+1}+1$ for some integer s, then

$$-1 \equiv p^{k^r s+1} \equiv p^{k((k^{r-1})s)+1} \equiv p^{kt+1} \mod q$$

when $t = (k^{r-1})s$ and we are done. Conversely, by contrapositive, suppose $q \nmid p^{k^r s+1}+1$, i.e., $q \nmid p^{k(k^{r-1}s)+1}+1$ for all integer $s \in \{0, 1, \ldots, \mathrm{ord}_q(p)\}$. Therefore $q \nmid p^{kt+1}+1$ for all integer $t = k^{r-1}s \in \{0, 1, \ldots, \mathrm{ord}_q(p)\}$ which is corresponding to s. $\square$

Corollary 31. Let p and k, then for all integer r, $\lambda_{p^{k^r},1}(q) = \lambda_{p^k,1}(q)$.

Theorem 32. Let p be prime and k be a positive integer having a canonical form $k = k_1^{l_1}k_2^{l_2}\ldots k_m^{l_m}$, where k_i's are prime and l_i's are positive integers.
Then $\lambda_{p^k,1}(q) = 0$ if and only if $\lambda_{p^{k_i},1}(q) = 0$ for all $i \in \{1, 2, \ldots, m\}$.

Proof. Suppose $\lambda_{p^k,1}(q) = 0$, then $q \mid p^{ks+1}+1$ for some integer s. Since $k_i \mid k$ for all $i \in \{1, 2, \ldots, m\}$, we have $q \mid p^{k_i \frac{k}{k_i}s+1}+1$ and thus $q \mid p^{k_i t_i+1}+1$ for an integer $t_i = \frac{k}{k_i}s+1$ for all i, that is, $\lambda_{p^{k_i},1}(q) = 0$. Conversely, by contrapositive, suppose $\lambda_{p^k,1}(q) = 1$, then $q \nmid p^{ks+1}+1$ for all integer s. Thus $q \nmid p^{k_i \frac{k}{k_i}s+1}+1$ for some i. Thus $\lambda_{p^{k_i},1}(q) = 1$. $\square$

Corollary 33. Let p be prime and k be a positive integer having a canonical form $k = k_1^{l_1}k_2^{l_2}\ldots k_m^{l_m}$, where k_i's are prime and l_i's are positive integers. Then $\lambda_{p^k,1}(q) = 1-(1-\lambda_{p^{k_1},1}(q))(1-\lambda_{p^{k_2},1}(q))\ldots(1-\lambda_{p^{k_m},1}(q))$.

Theorem 34. Let p be prime and k be a positive integer, then $\lambda_{p^k,1}(1) = 0$, $\lambda_{p^k,1}(2) = 0$ for all prime $p > 2$ and

$$\lambda_{p^k,1}(q) = \begin{cases} 0 & , 2 \mid \mathrm{ord}_q(p),\ p^{\frac{\mathrm{ord}_q(p)}{2}} = -1 \text{ and } \gcd(k, \frac{\mathrm{ord}_q(p)}{2}) = 1, \\ 1 & , \text{otherwise} \end{cases}$$

for all $q > 2$. Note that the calculation is taken on $\mathbb{Z}_q$.

Proof. Clearly, for all integer k, $\lambda_{p^k,1}(1) = 0$ since $1 \mid p^{k+1} + 1$ for all integer p and $\lambda_{p^k,1}(2) = 0$ since $2 \mid p^{\frac{\mathrm{ord}_q(p)}{2}k+1} + 1$ for all odd prime p.
Suppose $2 \mid \mathrm{ord}_q(p), p^{\frac{\mathrm{ord}_q(p)}{2}} = -1$ and $\gcd(k, \frac{\mathrm{ord}_q(p)}{2}) = 1$. There exists $m, n \in \mathbb{Z}$ satisfying $mk + n\frac{\mathrm{ord}_q(p)}{2} = 1$. We have that m, n are of the form $m = -d\frac{\mathrm{ord}_q(p)}{2} + a$ and $n = dk + b$ for some integers a, b, d such that $am + bn = 1$. We will show that n is odd for some choice of d. By contradiction, suppose that there is no $d \in \mathbb{Z}$ making n odd, i.e., n is even for all choices of d. Consider k in two cases. If k is odd, then we can choose d to be of the same parity to b which makes n even and contradicts the assumption. If k is even then $mk + n\frac{\mathrm{ord}_q(p)}{2}$ is even, a contradiction. Thus n is odd for some choice of d. Hence

$$-1 = (-1)^n = (p^{\frac{\mathrm{ord}_q(p)}{2}})^n = p^{-mk+1}$$

which is equivalent to $q \mid p^{-mk+1} + 1$. Therefore $\lambda_{p^k,1}(q) = 0$.

Conversely, if $2 \nmid \mathrm{ord}_q(p)$ then $p^x = -1$ has no solution, $p^{ks+1} = -1$ also does and $p^{\frac{\mathrm{ord}_q(p)}{2}} \neq -1$ gives the same result. Suppose $2 \mid \mathrm{ord}_q(p), p^{\frac{\mathrm{ord}_q(p)}{2}} = -1$ and $\gcd(k, \frac{\mathrm{ord}_q(p)}{2}) > 1$. We will show that $p^{kx+1} = -1$ has no solution. By contradiction, suppose there is, then $\frac{\mathrm{ord}_q(p)}{2} \mid kx + 1$. Since $\gcd(k, \frac{\mathrm{ord}_q(p)}{2}) \mid \frac{\mathrm{ord}_q(p)}{2}$, we have that $\gcd(k, \frac{\mathrm{ord}_q(p)}{2}) \mid kx+1$. So $\gcd(k, \frac{\mathrm{ord}_q(p)}{2}) \mid 1$ because $\gcd(k, \frac{\mathrm{ord}_q(p)}{2}) \mid k$, implying $\gcd(k, \frac{\mathrm{ord}_q(p)}{2}) = 1$ contradicts the assumption. So neither $p^{kx+1} = -1$ nor $q \mid p^{kx+1} + 1$ has no solution for all integer x therefore $\lambda_{p^k,1}(q) = 1$. □

Theorem 35. Let p be prime and k be a positive integer, then $\lambda_{p^k,T}(q) = 0$ if and only if $S_{p^k}(p^T) = S_{p^k}(-1)$ in $\mathbb{Z}_q$, in fact, if and only if $S_{p^k}(-p^T a) = S_{p^k}(a)$ in $\mathbb{Z}_m$ if and only if $S_{p^k}(a)$ is of type T.

Proof. Suppose $\lambda_{p^k,T}(q) = 0$, then by definition, $q \mid p^{ks+T} + 1$ for some integer $s \in \{0, 1, \ldots, \mathrm{ord}_q(p) - 1\}$. That is $p^{ks+T} = -1$, which is equivalent to, $-1 \in \{p^T, p^{k+T}, \ldots, p^{(\mathrm{ord}_q(p)-1)k+T}\}$, i.e., $S_{p^k}(p^T) = S_{p^k}(-1)$ in $\mathbb{Z}_q$. Conversely, suppose $S_{p^k}(p^T) = S_{p^k}(-1)$ in $\mathbb{Z}_q$, then $-1 = p^{ks+T}$ for some integer s and $q \mid p^{ks+T} + 1$ implying $\lambda_{p^k,T}(q) = 0$. □

Definition 36. Let a be a positive integer. If k is the greatest integer such that $2^k | a$ and $2^{k+1} \nmid a$, we write $2^k || a$.

Remark 37. For any positive integers a, b. If $a|b$, $2^k||a$ and $2^k||b$ then $\dfrac{b}{a}$ is odd.

Lemma 38. Let k, T and a be positive integers such that $T \leq k$. There exists an integer s such that $ks + T \equiv a \mod 2a$ if and only if $\gcd(k, a) \mid \gcd(k, T)$ and $\frac{\gcd(k,T)}{\gcd(k,a)}$ is odd.

Proof. Let x, y and z be integers such that $2^x||k$, $2^y||a$ and $2^z||T$. Suppose there exists an integer s such that $ks + T \equiv a \mod 2a$, thus $0 + T \equiv 0 \mod \gcd(k, a)$ hence $\gcd(k, a)|T$ which implies $\gcd(k, a)|\gcd(k, T)$ and the first part is done. Moreover, we have that $\min(x, y) \leq \min(x, z)$. It is clear that $2^{\min(x,y)}||\gcd(k, a)$ and $2^{\min(x,z)}||\gcd(k, T)$.

If $x \leq \min(y, z)$, then $x = \min(x, y) = \min(x, z)$ hence $2^x||\gcd(k, a)$ and $2^x||\gcd(k, T)$. Thus $\frac{\gcd(k,T)}{\gcd(k,a)}$ is odd.

If $x > \min(y, z)$, we have that $y \leq z$ so $y = \min(y, z) < x$. From $ks + T \equiv a \mod 2a$, it follows that

$$\frac{k}{2^y}s + \frac{T}{2^y} \equiv \frac{a}{2^y} \mod 2\frac{a}{2^y}.$$

Since $x > y$, $\frac{k}{2^y}$ is even and so is $\frac{k}{2^y}s$. Because $\frac{a}{2^y}$ is odd and $2\frac{a}{2^y}$ is even, we have that $\frac{T}{2^y}$ is odd thus $2^y||T$ which implies $y = z$ as we desired. So $\min(x, y) = \min(x, z)$ hence $2^{\min(x,y)}||\gcd(k, a)$ and $2^{\min(x,y)}||\gcd(k, T)$. Thus $\frac{\gcd(k,T)}{\gcd(k,a)}$ is odd.

Conversely, suppose $\gcd(k, a) \mid \gcd(k, T)$ and $\frac{\gcd(k,T)}{\gcd(k,a)}$ is odd. Clearly, $\gcd(k, a)|T$. We will show that there exists odd integer n such that $na \equiv T \mod k$. Since

$$T \equiv \frac{T\gcd(k, a)}{\gcd(k, a)} \equiv \frac{T(pa - qk)}{\gcd(k, a)} \equiv \frac{Tp}{\gcd(k, a)}a \mod k$$

for some integers p, q. If k is odd then either $n = \frac{Tp}{\gcd(k,a)}$ or $n = \frac{Tp}{\gcd(k,a)} + k$ is odd as we desired. Consider when k is even, we will show that $\frac{k+Tp}{\gcd(k,a)}$ is odd, i.e., $\frac{k}{\gcd(k,a)}$ and $\frac{Tp}{\gcd(k,a)}$ are having different parity. Note that $p = p_0 + \frac{k}{\gcd(k,a)}$ for some $0 \leq p_0 < \frac{k}{\gcd(k,a)}$. If $\frac{k}{\gcd(k,a)}$ is odd then p has a choice of even so that $\frac{Tp}{\gcd(k,a)}$ is even. If $\frac{k}{\gcd(k,a)}$ is even, it implies that $y < x$. From $\frac{\gcd(k,T)}{\gcd(k,a)}$ is odd, we have that $\min(x, z) = \min(x, y) = y$ so $y = z$ which implies $\frac{T}{\gcd(k,T)}$ is odd. Lastly we will show that p is odd, suppose p is even. From $2^y||\gcd(k, a)$, we have that

$\frac{\gcd(k,a)}{2^y} = p\frac{a}{2^y} - q\frac{k}{2^y}$ is odd. But p and $\frac{k}{2^y}$ are even, so do their linear combination which cause a contradiction. So p is odd and implies that $\frac{Tp}{\gcd(k,a)}$ is odd.

From $\frac{k}{\gcd(a,k)}a \equiv 0 \mod k$, $n = \frac{k+Tp}{\gcd(k,a)}$ is odd as we desired and satisfying $na \equiv T \mod k$, hence $ks = na - T$ for some integer s and

$$ks + T = na \equiv (2 \cdot \frac{n-1}{2} + 1)a \equiv \frac{n-1}{2} \cdot 2a + a \equiv a \mod 2a$$

since n is odd. This completes the proof. $\square$

Theorem 39. Let p be prime and k be a positive integer, then $\lambda_{p^k,T}(1) = 0$, $\lambda_{p^k,T}(2) = 0$ for all prime $p > 2$ and

$$\lambda_{p^k,T}(q) = \begin{cases} 0 & , 2 \mid \mathrm{ord}_q(p),\ p^{\frac{\mathrm{ord}_q(p)}{2}} = -1,\ \gcd(k, \frac{\mathrm{ord}_q(p)}{2}) \mid \gcd(k,T) \text{ and} \\ & \quad 2 \nmid \frac{\gcd(k,T)}{\gcd(k,\frac{\mathrm{ord}_q(p)}{2})}, \\ 1 & , \text{otherwise.} \end{cases}$$

for all $q > 2$ and $T < k$. Note that the calculation is taken on $\mathbb{Z}_q$.

Proof. Suppose $2 \mid \mathrm{ord}_q(p), p^{\frac{\mathrm{ord}_q(p)}{2}} = -1, \gcd(k, \frac{\mathrm{ord}_q(p)}{2}) \mid \gcd(k,T)$ and $2 \nmid \frac{\gcd(k,T)}{\gcd(k,\frac{\mathrm{ord}_q(p)}{2})}$. Then by lemma, we have that there exists an integer s such that $ks + T \equiv \frac{\mathrm{ord}_q(p)}{2} \mod \mathrm{ord_q(p)}$. Thus $-1 = p^{\frac{\mathrm{ord}_q(p)}{2}} = p^{ks+T}$ for some integer s and we are done since $q \mid p^{ks+T} + 1$ implies $\lambda_{p^k,T}(q) = 0$.

Conversely, suppose $2 \nmid \mathrm{ord}_q(p), p^{\frac{\mathrm{ord}_q(p)}{2}} \neq -1, \gcd(k, \frac{\mathrm{ord}_q(p)}{2}) \nmid \gcd(k,T)$ or $2 \mid \frac{\gcd(k,T)}{\gcd(k,\frac{\mathrm{ord}_q(p)}{2})}$. If $2 \nmid \mathrm{ord}_q(p)$ then $p^x = -1$ has no solution, in addition, also $p^{ks+T} = -1$ and $p^{\frac{\mathrm{ord}_q(p)}{2}} \neq -1$ gives the same result. The rest cases implies $ks + T \equiv \frac{\mathrm{ord}_q(p)}{2} \mod \mathrm{ord_q(p)}$ has no solution by the lemma, hence $p^{ks+T} = -1$ also has no solution. So $q \nmid p^{ks+T} + 1$ for all integer s, which implies $\lambda_{p^k,T}(q) = 1$. $\square$

Let's say wow!